

Zwergerl — hackathon brief

A competitive discovery game for Linz, scoped to a one-day hackathon that uses festival data and City of Linz open data.

Context for anyone picking this up: there is a much larger product spec behind this (v0.3, a 16-week multi-city platform plan). This document is deliberately not that. It is only the hackathon slice.

The idea in one sentence

Every hidden spot in Linz has one owner — and you take it by standing where they stood and reproducing their photograph.

How it works

- Walk to a spot. The app checks your GPS distance and makes you stand still for a few seconds.
- The camera unlocks and shows the compass heading the original photo was taken at.
- You reproduce the framing. The app verifies position, heading, and whether the photo actually matches.
- Match confirmed → you own the spot and earn points daily.
- Don't come back and your grip decays — then someone steals it.
- A live leaderboard makes it a rivalry rather than a checklist.

What the two data sources do

City of Linz open data (data.linz.gv.at / Open Commons Region Linz, CC-BY) supplies the candidate spots — monuments, fountains, public art, sculptures. The map seeds itself from a query instead of from months of manual curation.

Festival data creates *ephemeral gems*: stages, acts, and installations that exist only while the festival runs, then vanish permanently. Being there is the only way to have ever owned one. This is the mechanic the hackathon actually unlocks — it doesn't exist in the base product.

Where AI sits

Three distinct places, worth keeping separate:

1. **Verification (the core).** Comparing two photographs to decide they show the same subject from the same angle is a machine-learning problem — CLIP, run in-browser via `@xenova/transformers`. This is the differentiator and the riskiest piece.

2. **Content generation (cheap and reliable).** Open data gives a name and coordinates, nothing more. An LLM turns each row into a spoiler-free teaser plus a short story that unlocks after a successful claim. This is how a city seeds in minutes. Takes about an hour and will not fail.
3. **Build tooling.** Writing the code with Claude. True, but everyone in the room is doing it — not a slide.

If #1 dies, #2 still gives a genuine AI story in the demo.

The verification stack

Three independent gates, all of which must pass. Don't blend them into one score — a great image match should never excuse a 90° heading error.

Signal	What it proves	Rough tolerance
GPS distance	You were in the right place	≤ 40 m pass, 40–80 m review
GPS accuracy	The reading is trustworthy	reject reading above ~60 m, don't blame the user
Compass heading	You faced the right subject	$\leq 35^\circ$ pass, $\leq 60^\circ$ review
Dwell time	You actually stopped, didn't drive past	≥ 10 –20 s
CLIP similarity	The photo shows the right thing	calibrate against your own test photos

Order matters: check distance and heading first (instant), only run the model once those pass.

Why all three. Heading alone knows direction but not subject. CLIP alone knows subject but not where you stood — the same church from the opposite side still scores high. Together: right place, right direction, right subject.

Known gap: photographing a printed photo or a screen from the correct spot defeats all of it. The answer isn't technical — make winning boring. The prize is a leaderboard row, seasons reset.

Two bugs that will bite:

- Compass bearings wrap: 350° and 10° are 20° apart. $\text{Math.abs}(((a - b + 540) \% 360) - 180)$
- Browser geolocation requires HTTPS. Deploy early, not at hour 8.

- iOS needs `DeviceOrientationEvent.requestPermission()` called from inside a tap handler, and uses `webkitCompassHeading`; Android uses `alpha` with `absolute: true`. Handle both.
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The ~9 hour MVP

Split into a spine that must work and a spike that can be abandoned without regret.

Spine (hours 0-6) — non-negotiable

- Mobile web app. No auth; a name field in `localStorage`.
- ~10 spots from Linz open data plus 3 festival gems, imported once and hardcoded as JSON.
- Map with pins: grey free, coloured owned. A distance-sorted list is an acceptable substitute.
- Claim flow: distance check → dwell → heading check → capture photo → award points.
- Shared leaderboard (one Supabase table).

Spike (parallel, hours 3-7) — abandon at hour 7, decided now

- CLIP similarity in-browser. Precompute reference embeddings on a laptop, ship as JSON. Never ship reference *photos* — they're a cheat sheet.
- Test it first with 6 phone photos before building anything around it. If matched pairs don't score clearly above unmatched ones, you've learned the important thing in 30 minutes.

Hours 7-9 — protect these

- Go outside and claim 3 gems on real streets. This will surface bugs a laptop never shows.
- Get 3-5 people at the event to claim one. Their claims populate the demo leaderboard.
- Record a 2-minute backup video of a successful claim in good light. Venue wifi dies; indoor GPS won't settle. Every experienced demoer carries one.
- Rehearse the pitch once out loud.

Scope rule: if the spine isn't done by hour 6, stop adding anything. A working loop with 10 gems beats a broken loop with 50.

Success criterion for the day: one stranger, with no explanation, walks to a pin and successfully claims it.

Honest assessment

Strong: it has a mechanic, not just a screen. Most hackathon projects are a map, a chatbot, or a dashboard over the same data. This one asks someone to physically stand somewhere and

reproduce a stranger's framing. Physical ideas are rare in a room full of laptops.

Weak: the open-data fit is retrofitted — the concept predates the hackathon. Lead with the festival mechanic, which is honestly native to the brief.

The space is more crowded than it looks. Geocaching, Munzee, Turf, Pokémon Go all combine location, ownership, and points. Photo reproduction is a real differentiator, but "nobody has done this" is an overclaim that costs credibility if a judge knows the space.

The real fatal risk isn't technical, it's density. A dead map is a dead game, and that won't show in a demo. Have the answer ready: student networks, ESN, orientation week, a launch event that manufactures opening-night density in one district.

The best reason to do it isn't winning. It's that the whole product rests on one unanswered question — can a stranger reproduce someone else's framing well enough to verify? You can't answer that alone, because you remember standing there. A room full of strangers claiming your gems is exactly the experiment you need, and it's free.

Pitch lines

For the room: "Every hidden spot in Linz has one owner — and you take it by standing where they stood and reproducing their photograph." (*Pause after "one owner."*)

For a student: "It's a game where you steal your friends' favourite places in Linz by finding them."

For a mentor or funder: "On-device computer vision that verifies someone is physically standing where they claim to be. The game is how we collect the training data."

If the CLIP spike failed: "Presence is verified by GPS, heading and dwell time. Automated image matching is the next step — here's what I measured today about whether it's viable." That reads as engineering judgment, not as a gap.

Open questions

- Which specific Linz datasets map cleanly onto gem candidates? (Needs checking on the day.)
 - What format does the festival publish its programme in, and does it include coordinates per stage?
 - Is CLIP viable on a phone in-browser, or does the spike die? (Answer with 6 test photos before the event.)
 - Name: *Zwergerl* is warm and local, which is an asset in this specific room. It travels badly if the project continues.
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Assets already made

- `zwerger1-map-mockup.html` — standalone, no dependencies, opens on a phone. Five pin states, tappable.